

Nicholas Zhang

(604)-997-0377 | nicholas.zhang4008@gmail.com | Coquitlam, BC | github.com/NicholasZ4008 | nicholaszhang.ca

TECHNICAL SKILLS

Languages: C#, Java, JavaScript/TypeScript, Python, SQL, HTML/CSS

Frameworks & Tools: Unity, .NET/ASP.NET Core, React, Next.js, Node.js, Entity Framework Core, Jest, Docker, Git, Azure DevOps, PostgreSQL, SQL Server, Jira, Coolify

EXPERIENCE

Web Application Developer Co-op

May 2025 – Jan 2026

Public Services and Procurement Canada, Government of Canada

Vancouver, BC

- Built and maintained ASP.NET Core MVC applications with Entity Framework Core and SQL Server for federal maritime infrastructure, including a vessel booking system for the largest commercial drydock on the West Coast
- Implemented role-based access control and comprehensive form validation for booking workflows, coordinating requirements directly with client subject matter experts
- Refactored controllers and enhanced RESTful API endpoints for a hydrographic survey visualization system used to calculate safe navigation drafts on the Fraser River
- Built client-side interfaces for a maritime navigation reporting system (Avadepth) using vanilla JavaScript, implementing JSON-driven dynamic menus and date validation for tide and depth calculations

Project Manager and Web Developer Co-op

Jan 2025 – May 2025

Provincial Health Services Authority

Langley, BC

- Developed web applications using ASP.NET and React with TypeScript for an ongoing ERP integration project
- Wrote unit tests with Jest, improving code coverage and reducing post-release defects

Frontend Test Engineer Co-op

Jan 2023 – May 2023

Epic Safety Inc.

Vancouver, BC

- Developed Jest test suites for React components, increasing code coverage by 40%

Vice President of Technology

September 2023 – Present

The Forum SFU | theforumuniversity.com

Burnaby, BC

- Lead a development team building a responsive website (Next.js, Tailwind CSS) serving 100+ guests at events
- Implemented secure email signup with fastAPI backend and PostgreSQL, self-hosted on a personal coolify server

PROJECTS

Mon Battlers | *Unity, C#, ComfyUI*

March 2026 – Present

- Developing a Pokémon-inspired turn-based monster battler in Unity as a game development portfolio piece
- Architected a data-driven monster system using ScriptableObjects for species data and a plain C# `BattleMonster` class for runtime state, cleanly separating static and mutable concerns
- Implemented an FSM-based battle system (`PlayerTurn` → `EnemyTurn` → `ApplyEffects` → `CheckFaint`) with a type effectiveness matrix and damage formula

Digital Diversity Map | *JavaScript, PHP, SQL Server, Google Maps API*

Jan 2025 – May 2025

- Independently led full-stack development of an interactive web application for 500+ team members to share cultural connections, now used by the organization's Equity, Diversity & Inclusion team
- Integrated Google Maps API with Places Autocomplete, marker clustering, and real-time data persistence
- Followed WCAG 2.1 accessibility compliance with responsive design

An Arm And A Leg | *Unity, C#, Git*

April 2026

- Shipped a match-3 roguelike for Windows and macOS with an 8-person team, winning **Town One 2026**, a game jam hosted by the Quiver Games Incubator
- As one of three developers, mentored two junior teammates and contributed across the entire C# codebase — spanning the match-3 grid, combat resolution, and relic effect systems — pair-programming through complex bugs and reviewing pull requests under a tight jam deadline
- Coordinated a Git feature-branch workflow across programmers, artists, and designers, resolving Unity scene and prefab merge conflicts to keep the team unblocked

EDUCATION

Simon Fraser University

B.Sc. Computer Science

Vancouver, British Columbia

September 2026